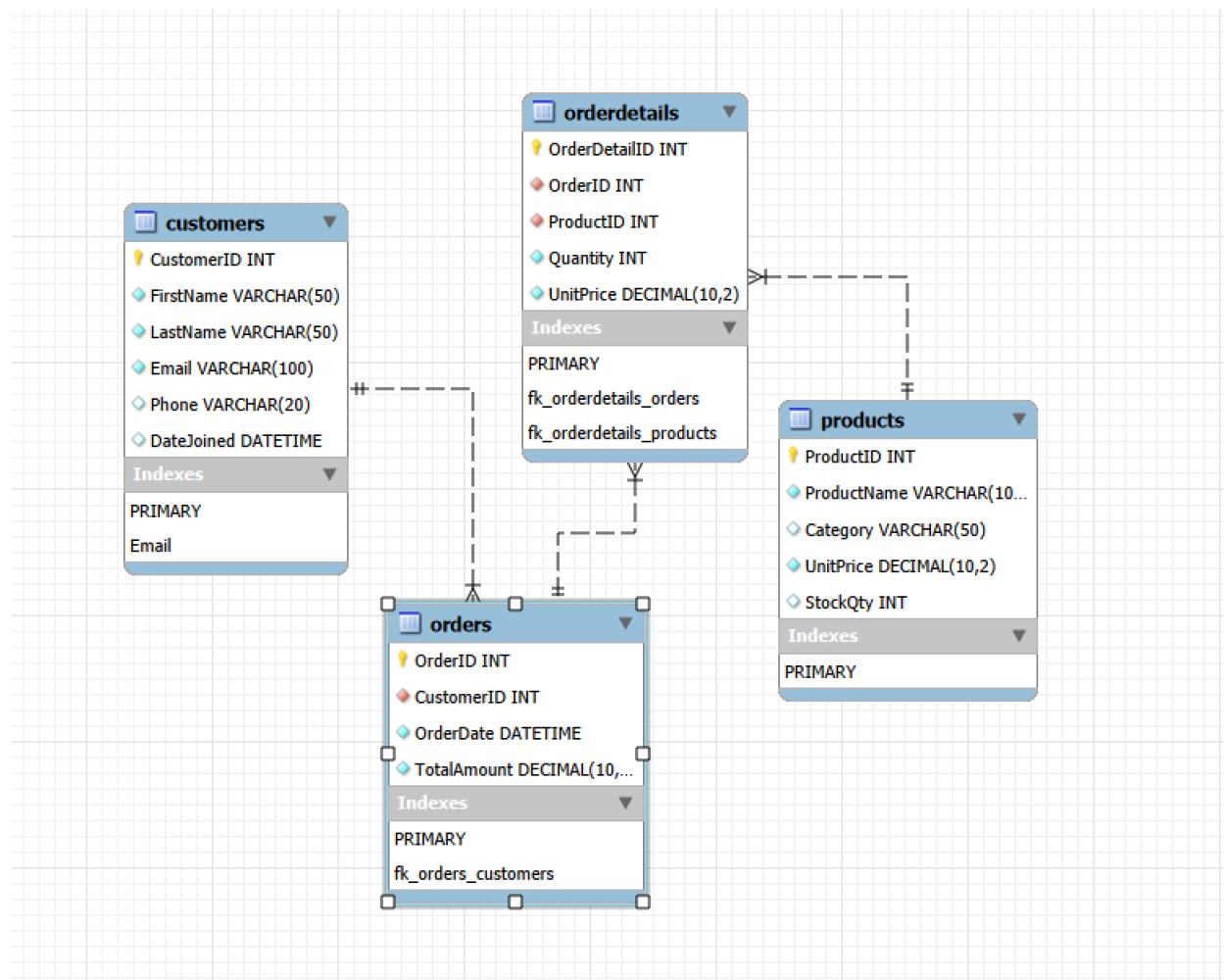


1. Database Modelling (5 Marks)

Design an E-Commerce Database Model

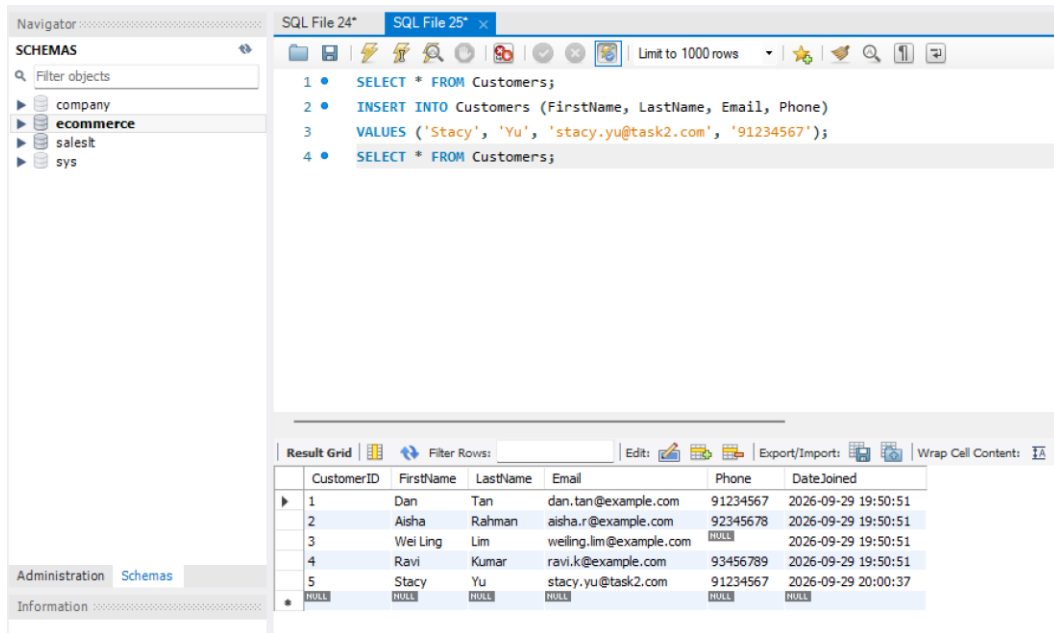
- Develop a data model suitable for an e-commerce application that includes at least three entities.
- Your model should clearly show the entity names and attributes for each table.
- Your model should clearly show primary keys and foreign keys, where appropriate.
- Your model should clearly show the relationships between tables (one-to-one, one-to-many, many-to-many), indicating how they connect and enforcing referential integrity.



2. Basic SQL Queries: Execute CRUD Operations (5 Marks)

Write SQL statements for the following tasks using hypothetical data from a typical e-commerce database:

- a) Insert a new customer record in the Customers table.



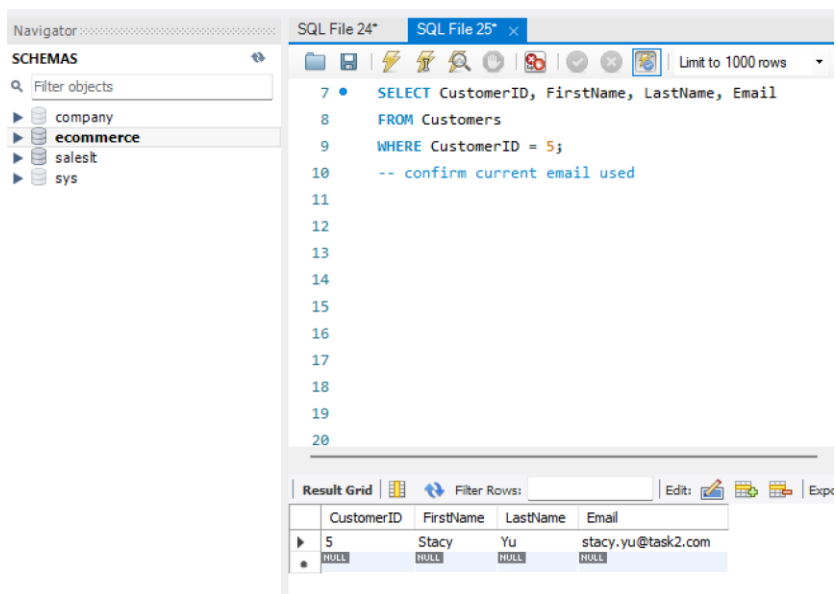
SQL File 24* SQL File 25* x

```
1 • SELECT * FROM Customers;
2 • INSERT INTO Customers (FirstName, LastName, Email, Phone)
3 • VALUES ('Stacy', 'Yu', 'stacy.yu@task2.com', '91234567');
4 • SELECT * FROM Customers;
```

Result Grid

CustomerID	FirstName	LastName	Email	Phone	DateJoined
1	Dan	Tan	dan.tan@example.com	91234567	2026-09-29 19:50:51
2	Aisha	Rahman	aisha.r@example.com	92345678	2026-09-29 19:50:51
3	Wei Ling	Lim	weiling.lim@example.com	NULL	2026-09-29 19:50:51
4	Ravi	Kumar	ravi.k@example.com	93456789	2026-09-29 19:50:51
5	Stacy	Yu	stacy.yu@task2.com	91234567	2026-09-29 20:00:37
•	NULL	NULL	NULL	NULL	NULL

- b) Update a customer's email address based on their unique CustomerID.



SQL File 24* SQL File 25* x

```
7 • SELECT CustomerID, FirstName, LastName, Email
8 • FROM Customers
9 • WHERE CustomerID = 5;
10 • -- confirm current email used
11
12
13
14
15
16
17
18
19
20
```

Result Grid

CustomerID	FirstName	LastName	Email
5	Stacy	Yu	stacy.yu@task2.com
•	NULL	NULL	NULL

Navigator: SQL File 24* SQL File 25* Limit to 1000 rows

Schemas: Filter objects

- company
- ecommerce
- saleslt
- sys

Administration Schemas

Information

Schema: ecommerce

```
7 • SELECT CustomerID, FirstName, LastName, Email
8 FROM Customers
9 WHERE CustomerID = 5;
10 -- confirm current email used
11 • UPDATE Customers
12 SET Email = 'yu.stacy@task2b.com'
13 WHERE CustomerID = 5;
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
```

Output

Action Output

#	Time	Action	Message
241	20:02:41	SELECT CustomerID, FirstName, LastName, Email FROM Customers WHERE CustomerID = 5 LIM...	1 row(s) returned
242	20:04:25	UPDATE Customers SET Email = 'yu.stacy@task2b.com' WHERE CustomerID = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

Object Info Session

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 24* SQL File 25* Limit to 1000 rows

Schemas: Filter objects

- company
- ecommerce
- saleslt
- sys

Administration Schemas

```
10 • SELECT CustomerID, FirstName, LastName, Email
11 FROM Customers
12 WHERE CustomerID = 5;
13 -- confirm current email used
14 • UPDATE Customers
15 SET Email = 'yu.stacy@task2b.com'
16 WHERE CustomerID = 5;
17 -- execute the update
18 • SELECT CustomerID, FirstName, LastName, Email
19 FROM Customers
20 WHERE CustomerID = 5;
21 -- confirm the row has been updated
22
23
```

Result Grid

CustomerID	FirstName	LastName	Email
5	Stacy	Yu	yu.stacy@task2b.com

c) Delete a specific record from the Orders table based on OrderID.

22

23 • `SELECT * FROM Orders;`

24 • `SELECT * FROM OrderDetails;`

25

26

27

Result Grid

OrderID	CustomerID	OrderDate	TotalAmount
1	1	2023-12-31 22:15:00	700.00
2	1	2024-01-05 10:30:00	520.00
3	2	2024-01-12 14:00:00	95.00
4	3	2024-01-20 09:45:00	500.00
5	2	2024-01-31 18:20:00	605.00
6	4	2024-02-01 00:00:00	700.00
7	3	2024-05-31 23:50:00	25.00
8	1	2024-06-03 11:00:00	210.00
9	4	2024-06-18 16:30:00	375.00
10	2	2024-06-30 21:05:00	280.00
11	3	2024-07-01 08:00:00	45.00
NULL	NULL	NULL	NULL

Result Grid

OrderDetailID	OrderID	ProductID	Quantity	UnitPrice
1	1	3	2	350.00
2	2	3	1	350.00
3	2	2	1	120.00
4	2	1	2	25.00
5	3	1	2	25.00
6	3	5	1	45.00
7	4	4	1	280.00
8	4	2	1	120.00
9	4	1	4	25.00
10	5	4	2	280.00
11	5	5	1	45.00
12	6	3	2	350.00
13	7	1	1	25.00
14	8	2	1	120.00
15	8	5	2	45.00
16	9	3	1	350.00
17	9	1	1	25.00
18	10	4	1	280.00
19	11	5	1	45.00
NULL	NULL	NULL	NULL	NULL

Output

#	Time	Action	Message	Duration / Fetch
247	20:09:38	SELECT * FROM OrderDetails LIMIT 0, 1000	19 row(s) returned	0.000 sec / 0.000 sec
248	20:11:19	DELETE FROM Orders WHERE OrderID = 11	1 row(s) affected	0.016 sec

23 • `SELECT * FROM Orders;`

24 • `SELECT * FROM OrderDetails;`

25 • `DELETE FROM Orders`

26 `WHERE OrderID = 11;`

27

Result Grid

OrderID	CustomerID	OrderDate	TotalAmount
1	1	2023-12-31 22:15:00	700.00
2	1	2024-01-05 10:30:00	520.00
3	2	2024-01-12 14:00:00	95.00
4	3	2024-01-20 09:45:00	500.00
5	2	2024-01-31 18:20:00	605.00
6	4	2024-02-01 00:00:00	700.00
7	3	2024-05-31 23:50:00	25.00
8	1	2024-06-03 11:00:00	210.00
9	4	2024-06-18 16:30:00	375.00
10	2	2024-06-30 21:05:00	280.00
NULL	NULL	NULL	NULL

Result Grid

OrderDetailID	OrderID	ProductID	Quantity	UnitPrice
10	5	4	2	280.00
11	5	5	1	45.00
12	6	3	2	350.00
13	7	1	1	25.00
14	8	2	1	120.00
15	8	5	2	45.00
16	9	3	1	350.00
17	9	1	1	25.00
18	10	4	1	280.00
19	11	5	1	45.00
NULL	NULL	NULL	NULL	NULL

```
27 • DELETE FROM OrderDetails
28 WHERE OrderID = 11;
29
30
31
```

[Context Help](#) [Snippets](#)

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 256	20:13:38	SELECT * FROM OrderDetails LIMIT 0, 1000	19 row(s) returned	0.000 sec / 0.000 sec
✓ 257	20:15:16	DELETE FROM OrderDetails WHERE OrderID = 11	1 row(s) affected	0.016 sec

⏮

```
23 • SELECT * FROM Orders;
24 • SELECT * FROM OrderDetails;
25 • DELETE FROM Orders
```

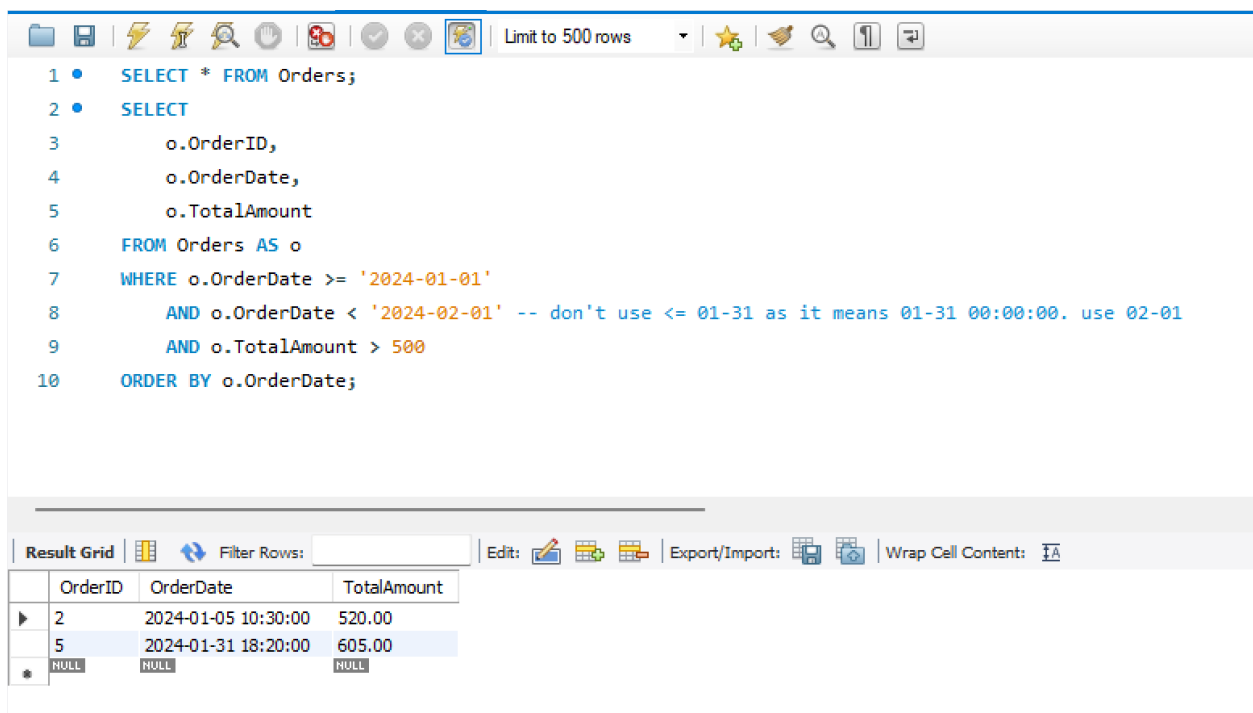
Result Grid   Filter Rows: Edit:    Export/Import:   W

	OrderDetailID	OrderID	ProductID	Quantity	UnitPrice
	10	5	4	2	280.00
	11	5	5	1	45.00
	12	6	3	2	350.00
	13	7	1	1	25.00
	14	8	2	1	120.00
	15	8	5	2	45.00
	16	9	3	1	350.00
	17	9	1	1	25.00
	18	10	4	1	280.00
*	NULL	NULL	NULL	NULL	NULL

3. Advanced SELECT Queries: Filtered Data Retrieval (5 Marks)

Write a SELECT query to fetch all orders from the Orders table that meet the following criteria:

- a) Orders were placed in January 2024.
- b) The order total exceeds \$500.
- c) Your query should only include columns for OrderID, OrderDate, and TotalAmount in the result.



The screenshot displays a SQL query editor interface. The top toolbar includes icons for file operations, execution, and a 'Limit to 500 rows' dropdown. The query text is as follows:

```
1 • SELECT * FROM Orders;
2 • SELECT
3     o.OrderID,
4     o.OrderDate,
5     o.TotalAmount
6 FROM Orders AS o
7 WHERE o.OrderDate >= '2024-01-01'
8       AND o.OrderDate < '2024-02-01' -- don't use <= 01-31 as it means 01-31 00:00:00. use 02-01
9       AND o.TotalAmount > 500
10 ORDER BY o.OrderDate;
```

Below the editor is the 'Result Grid' section, which includes a 'Filter Rows' input field and buttons for 'Edit', 'Export/Import', and 'Wrap Cell Content'. The grid shows the following data:

	OrderID	OrderDate	TotalAmount
▶	2	2024-01-05 10:30:00	520.00
	5	2024-01-31 18:20:00	605.00
*	NULL	NULL	NULL

4. SQL Joins for Data Aggregation (5 Marks)

Retrieve Information Using Joins

a) Using the tables Products, Orders, and OrderDetails, write an SQL query.

b) The query should retrieve information on products ordered in June 2024.

c) The result should display:

ProductName: The name of the product.

QuantityOrdered: The quantity of each product in the order.

OrderDate: The date the order was placed.

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'SCHEMAS' pane displays a tree view with 'ecommerce' selected. The main pane shows a SQL query in 'SQL File 27*'. The query is as follows:

```
1 • SELECT * FROM Orders;
2 • SELECT * FROM OrderDetails;
3 • SELECT * FROM Products;
4 • SELECT
5     p.ProductName,
6     od.Quantity AS QuantityOrdered,
7     DATE_FORMAT(o.OrderDate, '%Y-%m-%d') AS OrderDate
8 FROM Orders AS o
9     JOIN OrderDetails AS od
10        ON o.OrderID = od.OrderID
11     JOIN Products AS p
12        ON od.ProductID = p.ProductID
13 WHERE o.OrderDate >= '2024-06-01'
14        AND o.OrderDate < '2024-07-01'
15 ORDER BY o.OrderDate, p.ProductName;
```

Below the query, the 'Result Grid' shows the following data:

ProductName	QuantityOrdered	OrderDate
Laptop Stand	2	2024-06-03
Mechanical Keyboard	1	2024-06-03
27-inch Monitor	1	2024-06-18
Wireless Mouse	1	2024-06-18
Office Chair	1	2024-06-30